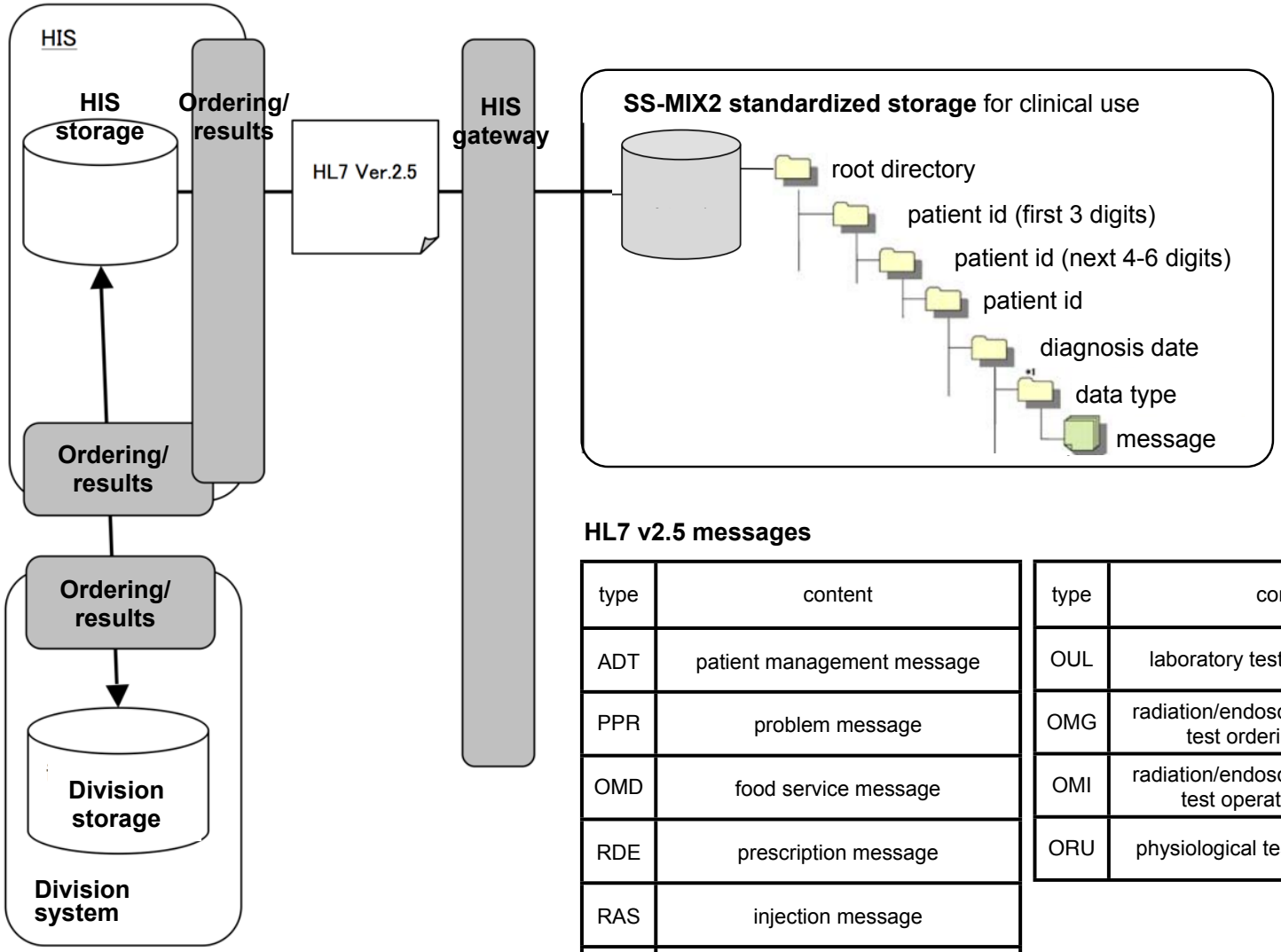
SS-MIX2 standardized storage for clinical use

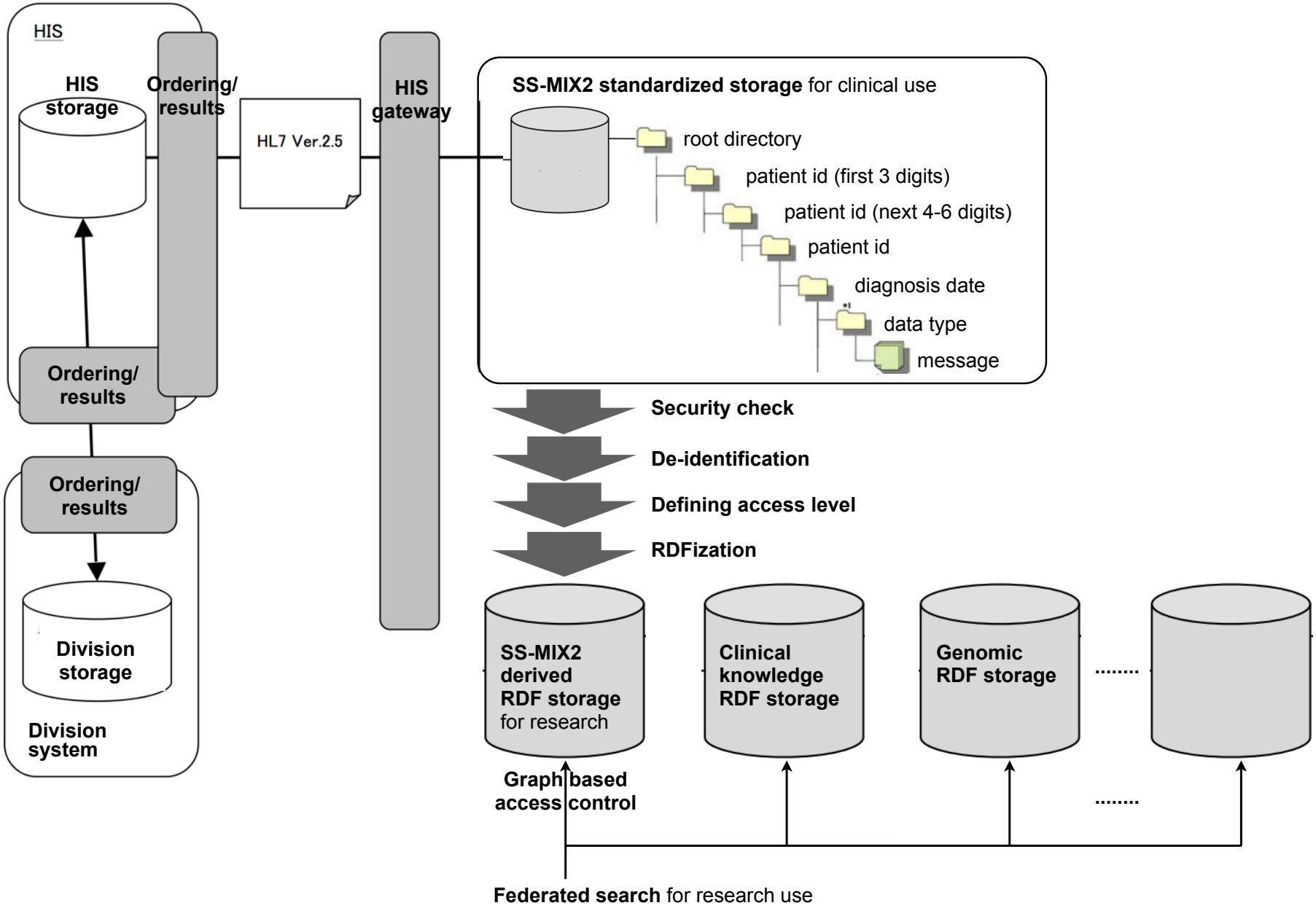


HL7 v2.5 messages

type	content
ADT	patient management message
PPR	problem message
OMD	food service message
RDE	prescription message
RAS	injection message
OML	laboratory test message

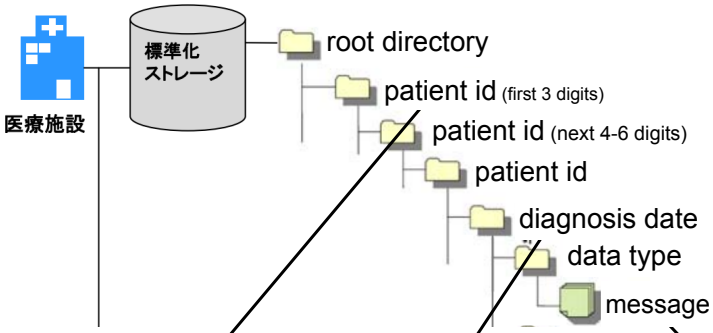
type	content
OUL	laboratory test result message
OMG	radiation/endoscope/physiological test ordering message
OMI	radiation/endoscope/physiological test operation message
ORU	physiological test result message

SS-MIX2 RDF storage for research use



HL7 v2.5 message RDF model

SS-MIX2 standardized storage



patient id	diagnosis date
0001	09/12/2011

HL7 v2.5 message PPR (problem) message

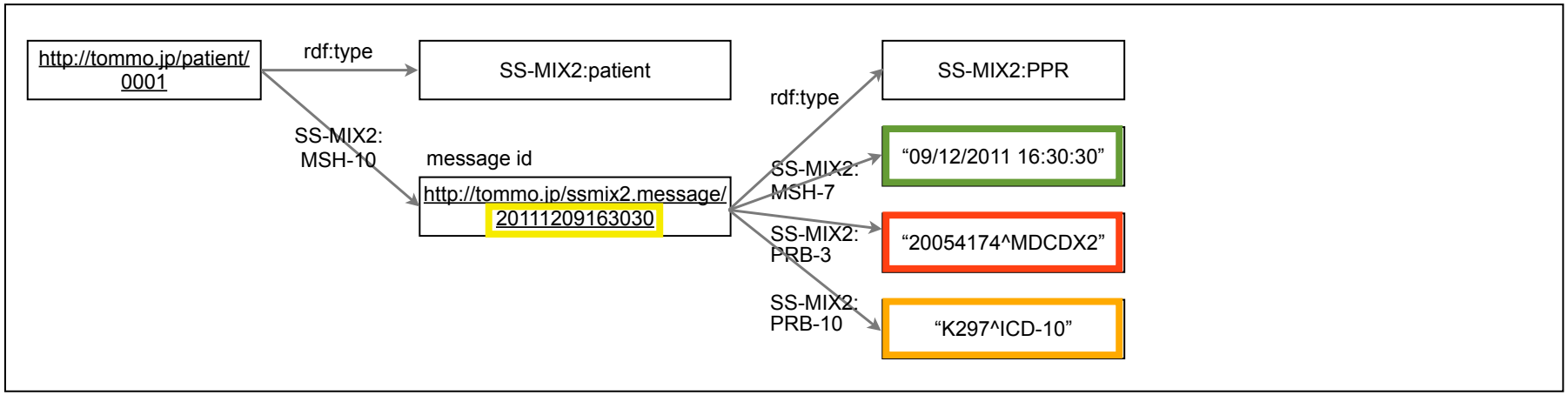
病名(歴)情報登録・更新(PPR^ZD1)

```
MSH|^~¥&|HIS123|SEND|GW|RCV|20111209163030||PPR^ZD1^PPR_ZD1|201112091630305|P|2.5||||||~ISO IR87||ISO
 2022-1994
PID|0001||1234567890^^^^PI||患者^太郎^^^^^L^I~カンジャ^タロウ^^^^^L^P||19650415|M
PRB|AD|20110915|20054174^胃炎^MDCDX2|123456789012345|||20110831||20110915|K297^^I10^0^外来時
 ^JHSD0004||||N^回復せず^HL70241|20110915|20110831|胃炎|1^主診断^JHSD0007||||||V^非常に限定^HL70177
ZPR||20054174^胃炎^MDCDX2|||TSQF^胃炎^MDCDX2
ZI1|1|01^全国健康保険協会管掌健康保険^JHSD0001|1130012|全国健康保険協会東京支部||||||123456|11010203
 |20110901|20120831||13||SEL^本人^HL70063|19650415|^^^^1050001^^H^東京都港区虎ノ門１－１９－９||70
ORC|NW|123456789012345||||||||||110^医師^一郎^^^^^^^L^^^^^I^^イシ^イチロウ^^^^^^^L^^^^^P|||||01^内科
 ^MML028||||||||||||0^外来患者オーダ^HL70482
```

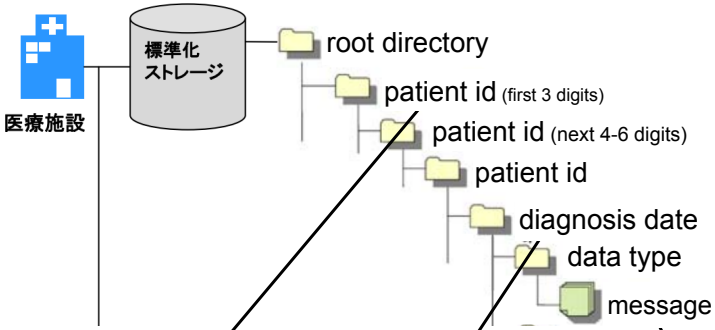
messeage type	segment type	field	value	access lv
PPR	MSH	MSH-10 (message ID)	“20111209163030”	“A01”
	PRB	PRB-3 (diagnosed disease code)	“20054174^MDCDX2”	
	PRB	PRB-10 (diagnosed disease code)	“K297^ICD-10”	

MSH: message segment
PRB: problem detailed segment

HL7 v2.5 RDF graph for access_lv A01 PPR (problem) message



SS-MIX2 standardized storage



patient id	diagnosis date
0001	09/12/2011

HL7 v2.5 message

OUL (laboratory test result) message

検体検査結果通知(OUL^R22)

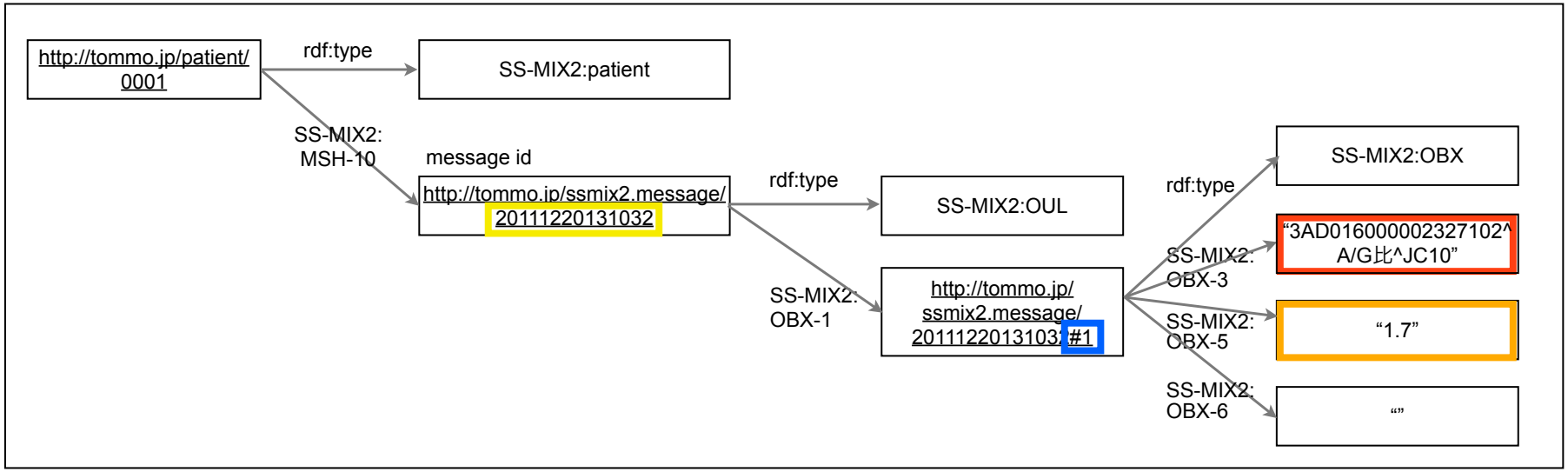
```
MSH|^~¥&|HIS123|SEND|GW|RCV|20111220103059.0000||OUL^R22^OUL_R22|20111220131032|P|2.5||||||~ISO IR87||ISO
 2022-1994
PID|0001||0001000052^^^^PI||患者^太郎^^^^^L^I~カンジャ^タロウ^^^^^L^P||19700405|M|||^^^^422-8033^JPN^H^静
 岡県静岡市登呂１－３－５
PV1|0001|O|01^^^^^C
SPM|1|000000001219001||023^血清^JC10||||||||||||201112191500
OBR|1|000000011000354|000000001219001|E001^血液学的検査^99003|||20111219|20111219|||||||607^医師一郎
 ^^^^^^^^L^^^^^I||||||20111220103059
ORC|SC|000000011000354|000000001219001||CM||||20111220103059|||607^医師一郎^^^^^^^^L^^^^^I|01^^^^^C|||01^
 内科^99XY1||||登呂病院|||||||0^外来患者オーダ^HL70482
OBX|1|NM|3A016000002327102^A/G比^JC10||1.7||1.2-2.0||||F|||201112191500
OBX|2|NM|3A010000002327101^総蛋白^JC10||7.2|g/dl^g/dl^99XYZ|6.70-8.3||||F|||201112191500
OBX|3|NM|3A015000002327101^アルブミン^JC10||4.9|g/dl^g/dl^99XYZ|3.7-5.5||||F|||201112191500
```

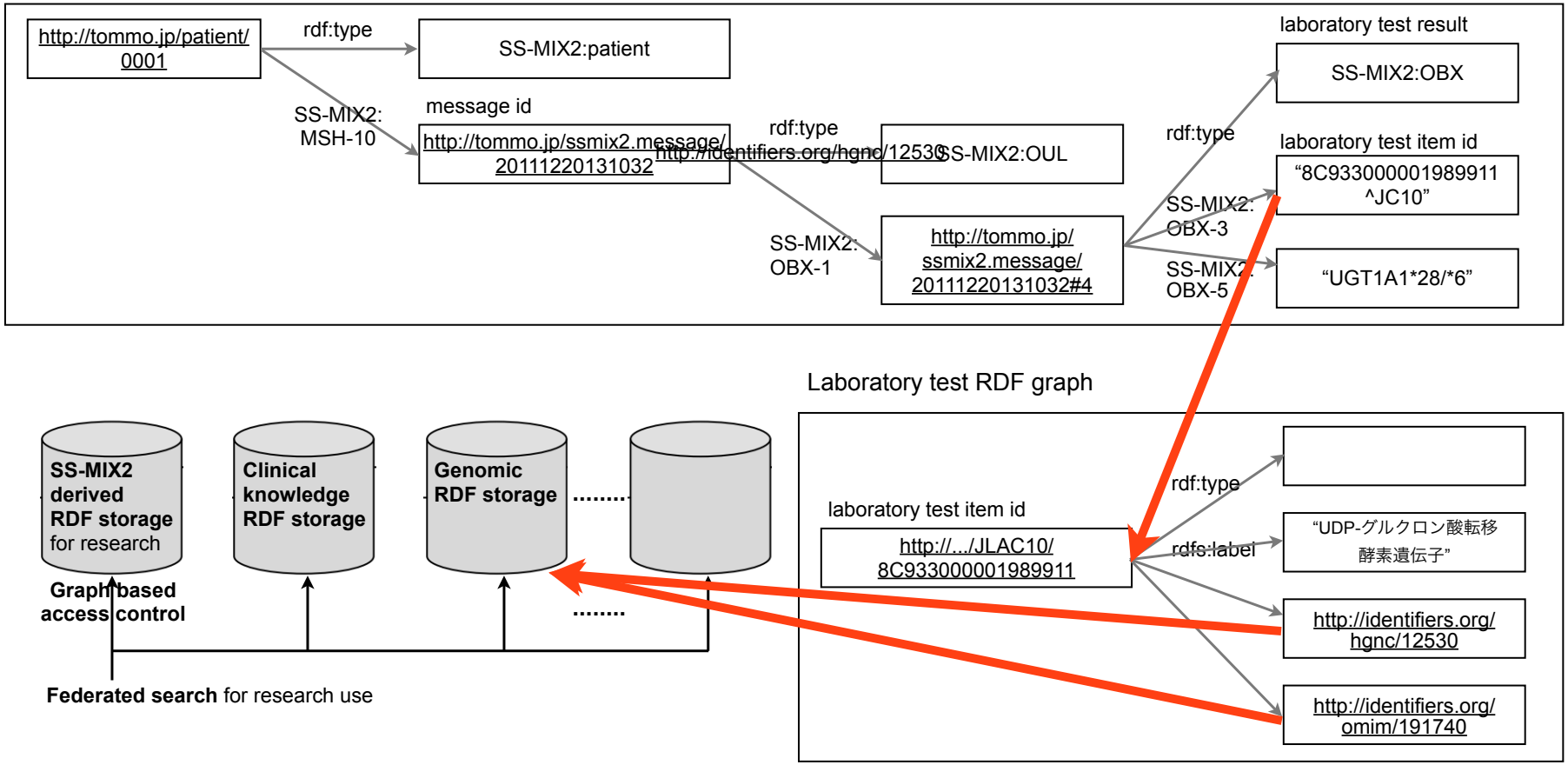
messeage type	segment type	field	value	access lv
OUL	OBX	OBX-1 (set ID - OBX)	“1”	“A01”
	OBX	OBX-3 (test item code)	“3AD016000002327102^A/G比^JC10”	
	OBX	OBX-5 (test result value)	“1.7”	

MSH: message segment
OBX: test result segment

HL7 v2.5 RDF graph for access_lv A01

OUL (laboratory test result) message



Use case: utilizing both HL7 v2.5 message RDF and genomic RDF

HL7 v2.5 RDF graph for access_lv A01 OUL (laboratory test result) message